

Definition 1: The integer n is even if there exists an integer k such that $n=2k$, and n is odd if there exists an integer k such that $n=2k+1$. (Note that every integer is either even or odd, and no integer is both even and odd.) Two integers have the same parity when both are even or both are odd; they have opposite parity when one is even and the other is odd. $\square$

Example 1: Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Soln: Let n be an arbitrary odd integer. By defn 1, there exists an integer k s.t. $n=2k+1$.

$$\therefore n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2p + 1 \text{ where}$$

$$p = 2k^2 + 2k. \text{ It's easy to see that } p \text{ is an integer. (From closure of real no.s)}$$

By defn 1, n^2 is an odd integer. $\square$

Example 2: Give a direct proof that if m and n are both perfect squares, then nm is also a perfect square. (An integer a is a perfect square if there is an integer b such that $a=b^2$.)

Soln: Let m and n be both perfect squares. By defn. of perfect square, there exists two integers x and y respectively s.t.

$$m=x^2 \text{ and } n=y^2.$$

$$\therefore nm = y^2 x^2 = (y \cdot y) \cdot (x \cdot x) = y \cdot (y \cdot (x \cdot x)) \text{ (Associativity of mult.)}$$

$$= y \cdot ((y \cdot x) \cdot x) \text{ (Associativity of mult.)}$$

$$= (y \cdot x) \cdot (y \cdot x) \text{ (Associativity of mult.)}$$

$$= (y \cdot x)^2 \quad \because y \cdot x \text{ is an integer, by defn. of perfect square, } nm \text{ is a perfect square. } \square$$